

Fluid Antenna System-Assisted Physical Layer Secret Key Generation

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Abstract—This paper investigates physical-layer key generation (PLKG) in multi-antenna base station systems, by leveraging a fluid antenna system (FAS) to dynamically customize radio environments. Without requiring additional nodes or extensive radio frequency (RF) chains, the FAS effectively enables adaptive antenna port selection by exploiting channel spatial correlation to enhance the secret key rate (SKR) at legitimate nodes. To comprehensively evaluate the performance of the FAS in PLKG, we propose an FAS-assisted PLKG model that integrates transmit beamforming and sparse port selection under independent and identically distributed (i.i.d.) and spatially correlated channel models, respectively. Specifically, the PLKG utilizes reciprocal channel probing to derive an approximate SKR expression based on the mutual information between legitimate channel estimates, explicitly accounting for the Eve's channel observation under spatially correlated channel scenarios. Nonconvex optimization problems for these scenarios are formulated to maximize the SKR subject to transmit power constraints and sparse port activation. We propose an iterative algorithm by capitalizing on successive convex approximation and Cauchy-Schwarz inequality to obtain a locally optimal solution. A reweighted ℓ_1 -norm-based algorithm is applied to advocate for the sparse port activation of FAS-assisted PLKG. To approximate the optimal activated ports obtained by exhaustive search, a low-complexity sliding window-based port selection is proposed to substitute reweighted ℓ_1 -norm method based on Rayleigh-quotient analysis. Simulation results demonstrate that the FAS-assisted PLKG scheme significantly outperforms fixed antenna-assisted PLKG schemes in both environments. It is shown that the FAS achieves higher SKR with fewer RF chains through dynamic sparse port

selection, which effectively reduces the resource overhead. Also, the sliding window approach closely approximates the globally optimal port selection compared to the reweighted ℓ_1 -norm method, rendering it suitable for practical deployments.

Index Terms—Physical layer security, secret key generation, fluid antenna system, spatially correlated channels.

I. INTRODUCTION

THE exponential growth of wireless communication systems demands robust security frameworks to protect sensitive data over shared wireless channels [1]. In dynamic and decentralized networks, such as the Internet-of-Things (IoT), traditional cryptography relying on pre-distributed keys faces critical limitations such as constrained computational resources, frequent topology variations, and impractical centralized key management [2]. To address these challenges, physical-layer key generation (PLKG) leverages inherent wireless channel characteristics to establish shared secret keys without pre-existing infrastructure [3]. By extracting entropy from channel state information [4], [5], [6], the PLKG enables legitimate transceivers to generate correlated keys via their common propagation environment, while eavesdroppers at different locations observe decorrelated channels [7]. This spatial decorrelation underpins the foundation of information-theoretic security, preventing reliable key estimation even for adversaries with unbounded computing power [8], [9].

Despite these advantages, PLKG remains challenging due to inherent limitations associated with wireless propagation and adversarial environments. Specifically, the performance of PLKG heavily relies on channel reciprocity and dynamic fading characteristics to extract the secure randomness [10]. However, practical implementations face significant obstacles in scenarios involving shadowed environments or static channels, where insufficient channel entropy undermines reliable key generation [11]. Additionally, directional beamforming introduces spatial sparsity that further reduces the available randomness [12]. Moreover, conventional approaches struggle to maintain security in scenarios with correlated eavesdropping channels or devices with limited resources, where pre-shared keys are impractical [13]. Recent advancements have mitigated these challenges through innovative architectural and signal processing techniques. Particularly, in traditional multiple fixed-antenna (FA) communication systems, beam-domain channel modeling leverages spatial

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decorrelation to isolate legitimate users from eavesdroppers. This approach significantly reduces pilot overhead and enhances security by exploiting sparse channel representations in the beamspace [14]. Furthermore, combining hybrid beamforming with channel estimates enables systems to dynamically adapt, compensating for any potential reciprocity mismatches and improving key-generation robustness [15]. However, FA-based approaches typically require numerous radio frequency (RF) chains, thereby increasing deployment complexity and communication overhead.

To overcome these challenges, a fluid antenna system (FAS) has been introduced to dynamically reconfigure antenna positions in practical scenarios, thereby effectively exploiting spatial diversity as a versatile alternative to traditional fixed antenna arrays [16], [17], [18]. Specifically, by adaptively selecting optimal ports via software control, the FAS enhances channel gains and achieves improved performance with a reduced number of RF chains [19]. This agility effectively lowers hardware complexity and energy consumption compared to fixed arrays, positioning the FAS as an ideal solution for enabling next-generation wireless networks [20], [21]. Owing to its tremendous potential, the notion of FAS has been applied to physical-layer secure communications through adaptive antenna positioning [22]. Meanwhile, the efficient integration of FAS with other technologies such as reconfigurable intelligent surface [23], [24] and non-orthogonal multiple access [25] has been investigated to further enhance secrecy performance. Recent studies have demonstrated the excellent ability of the FAS to maximize the secrecy rate by optimizing antenna positions to strengthen legitimate channels while disrupting eavesdroppers [26]. For instance, coding-enhanced cooperative jamming in the FAS has been shown to outperform conventional Gaussian noise jamming by exploiting spatial diversity to effectively harness and manipulate interference [27]. Additionally, FAS-assisted systems achieve robust secrecy performance under correlated fading channels, highlighting the promise for secure wireless communications [28].

On the other hand, in wireless IoT networks, PLKG is tailored for resource-constrained devices. However, traditional PLKG designs often demand numerous RF chains, overburdening these devices' hardware resources [9], [29], [30]. Motivated by the above studies on FAS, the integration of FAS and PLKG potentially addresses a limitation in lightweight IoT deployments. Indeed, employing the FAS can address this issue by dynamically reconfiguring activated ports with fewer RF chains [31], [32], [33]. In particular, with the requirement for few RF chains, the dynamic spatial reconfiguration of FAS is expected to increase the entropy of key extraction at the legitimate node. In addition, the larger number of potential FAS preset ports also introduces additional randomness to the PLKG compared to traditional fixed antennas [34]. Nonetheless, existing multi-antenna PLKG strategies, e.g., [12] and [13], require computationally intensive signal processing in the beam domain to improve the secret key rate (SKR). In addition, current security strategies for FAS, e.g., [26], [27], and [28], primarily consider Wyner-based degenerate eavesdropping channel assumptions and neglect secure communication

scenarios where the eavesdropper's channel may outperform that of the legitimate receiver. References [26], [27], and [28] characterize the upper bound of FAS-enabled secrecy rate but do not establish any performance bound for SKR. In contrast, the performance upper bound of fixed antenna-assisted PLKG in [12] and [13] scales directly with the number of RF chains (i.e., more RF chains lead to a higher SKR), but does not identify the performance limit for achieving the highest SKR with the minimum number of RF chains. Thus, the impact of FAS on PLKG still remains unexplored. As a matter of fact, addressing this gap requires developing sophisticated modeling techniques and deriving accurate expression for the secret key rate applicable to various scenarios. Moreover, it is imperative to surmount the obstacles inherent in non-convex optimization, particularly related to sparse port selection and beamforming design. Optimal port selection is a combinatorial problem with a ℓ_0 -norm sparsity, which is computationally intractable for the problem. These challenges necessitate innovative solutions that bridge information theory and optimization tailored to efficient secret key generation.

To investigate FAS-assisted PLKG in a multiple-input-single-output (MISO) system, this paper considers both i.i.d. and spatially correlated channel scenarios. Our main contributions are summarized as follows.

- We propose two FAS-assisted PLKG models based on an independent scenario, where Eve's estimate is independent of the legitimate users' estimates, and a correlated scenario that incorporates spatial correlation between eavesdropping and legitimate channels. Both scenarios are analyzed by leveraging mutual information and joint entropy analysis to derive approximate SKR expressions. We explicitly incorporate the FAS port activation matrix and spatial correlation structure into the SKR analysis and quantify how dynamic port selection enhances key-generation performance by strengthening the mutual information of legitimate channels while suppressing the correlation observable at the eavesdropper. Additionally, our analysis shows that the SKR gain increases with the largest eigenvalue of the spatially correlated matrix at the base station (BS).
- To maximize a tractable surrogate function, the nonconvex terms in the formulated problem are addressed via successive convex approximation (SCA) combined with the Cauchy-Schwarz inequality relaxation. To promote sparsity in the solution to port activation problem, we propose a reweighted ℓ_1 -norm method. Furthermore, by leveraging Rayleigh-quotient theory and the oscillatory decaying property of Bessel-function-based spatial correlation, we propose a low-complexity and near-optimal port-selection framework. To approximate the optimal activated ports obtained by exhaustive search, a sliding window-based port selection is proposed to improve port activation guided by Rayleigh-quotient theory, significantly reducing computational complexity.
- Through simulation validation, we reveal the performance of FAS-assisted PLKG, which serves as an implementation boundary for achieving a high SKR with a minimal number of RF chains. Simulation results demonstrate that

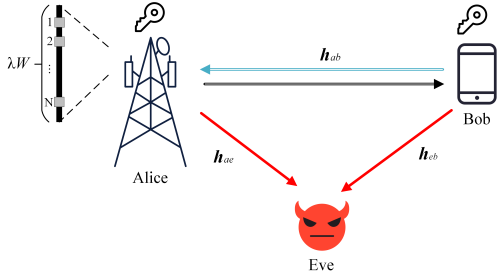


Fig. 1. The model of FAS-assisted secret key generation in MISO systems.

the proposed FAS-assisted PLKG scheme significantly outperforms the fixed antenna-assisted PLKG scheme in both independent and spatially correlated scenarios. With the same order of complexity, the sliding window-based port selection method consistently achieves superior SKR performance, compared to the reweighted ℓ_1 -norm method. In addition, the sliding window approach closely approximates the optimal port selection obtained by traversing all ports with exceedingly high computational complexity.

Notations: In this paper, matrices and vectors are denoted by boldface upper-case and boldface lower-case, respectively. $\mathbb{C}^{A \times B}$ denotes the space of complex matrices of size $A \times B$. $\mathcal{O}(\cdot)$ denotes the complexity. $(\cdot)^*$, $(\cdot)^\top$, and $(\cdot)^H$ denote the conjugate, transpose, and conjugate transpose, respectively. $\text{diag}(\mathbf{x})$ is a matrix whose main diagonal elements are the entries of $\mathbf{x}$. $\frac{df(x)}{dx}$ denotes that the function $f(x)$ takes the derivative of x . $\det(\cdot)$ is the matrix determinant. $\|\mathbf{x}\|_0$, $\|\mathbf{x}\|_1$, and $\|\mathbf{x}\|_2$ denote the ℓ_0 , ℓ_1 , and ℓ_2 norms of vector $\mathbf{x}$, respectively. $\mathbb{E}\{\cdot\}$ represents statistical expectation. $\lambda_{\max}(\mathbf{X})$ is the maximum eigenvalue of matrix $\mathbf{X}$. $\mathbf{b} \sim \mathcal{CN}(\mathbf{0}, \Sigma)$ denotes that $\mathbf{b}$ is a circularly symmetric complex Gaussian (CSCG) vector with zero mean and covariance matrix Σ . $\text{Cov}[a, b]$ denotes the covariance of random variables a and b . $\mathcal{I}(X | Y)$ is the mutual information between X and Y . $J_0(\cdot)$ is the zero-order Bessel function of the first kind. $\text{Re}\{\cdot\}$ represents the real part of a complex number.

II. SYSTEM MODEL

As shown in Fig. 1, we investigate a MISO system implementing an FAS-assisted PLKG scheme. Assuming a time-division duplexing (TDD) protocol is adopted, a multi-antenna BS, Alice, and a single antenna user, Bob, aim to generate symmetric keys by exploiting the reciprocity of the wireless channel with the assistance of an FAS. Meanwhile, a single antenna eavesdropper, Eve, attempts to intercept and decode the secret key information contained within the received signals¹.

A. Channel Model

In the FAS-assisted PLKG, we assume that Alice is equipped with a linear fluid antenna surface with N RF

¹Note that the considered framework can be generalized to multiple single-antenna eavesdroppers by optimizing against the worst case scenario (i.e., the eavesdropper with the strongest channel conditions).

chains, which is regarded as a common case of FAS [34]. M predetermined ports share N RF chains and are uniformly distributed along a linear space of length λW , where λ is the signal carrier wavelength and W is the normalized size of FAS. The N RF chains can adaptively alter their activated ports among the M available pre-set ports such that the PLKG performance can be enhanced.² Compared with the traditional MISO systems, the port switching capability of FAS offers more spatial DoF to the PLKG [34]. Since the spatial correlation affects the secret key rate, we consider the general spatial correlation channel model at Alice. The direct channels of Alice-to-Bob, Eve-to-Bob, and Alice-to-Eve are denoted by $\mathbf{h}_{ab} \in \mathbb{C}^{M \times 1}$, $\mathbf{h}_{eb} \in \mathbb{C}^{1 \times 1}$, and $\mathbf{h}_{ae} \in \mathbb{C}^{M \times 1}$, respectively.³ To account for the spatial correlation,⁴ the channel matrices are described by employing the FAS correlation channel model [31] as $\mathbf{h}_{ab} = \beta_{ab}^{\frac{1}{2}} \mathbf{J}_A^{\frac{1}{2}} \tilde{\mathbf{h}}_{ab}$, $\mathbf{h}_{ae} = \beta_{ae}^{\frac{1}{2}} \mathbf{J}_A^{\frac{1}{2}} \tilde{\mathbf{h}}_{ae}$ respectively, where $\mathbf{J}_A \in \mathbb{C}^{M \times M}$ is the spatial correlation matrix at Alice [36], [40]. The (n, m) -th element of $\mathbf{J}_A$ can be expressed as

$$\mathbf{J}_A(n, m) = \text{Cov}[h_n, h_m] = J_0\left(2\pi \frac{|n-m|}{M-1} W\right). \quad (1)$$

In addition, $\tilde{\mathbf{h}}_{ab} \in \mathbb{C}^{M \times 1}$ and $\tilde{\mathbf{h}}_{ae} \in \mathbb{C}^{M \times 1}$ are random vectors with i.i.d. Gaussian random entries of zero mean and unit variance. β_{ab} and β_{ae} are the path losses of the corresponding channels.

B. PLKG System Based on Transmit Beamforming

In this section, we propose a PLKG framework based on the established channel model of FAS. In the considered FAS-assisted PLKG system, Alice and Bob acquire reciprocal channel estimates by performing channel probing. The detailed procedure is introduced as follows [37], [41].

In the downlink stage, initiated by Alice's transmission of a predefined pilot sequence $s_d \in \mathbb{C}$ with unit power ($|s_d|^2 = 1$),⁵ the received signals at Bob and Eve are modeled as:

$$y_l^d = \mathbf{h}_{al}^\top \mathbf{w} s_d + z_l^d, l \in \{b, e\}, \quad (2)$$

where $\mathbf{w} \in \mathbb{C}^{M \times 1}$ denotes the sparse beamforming vector at Alice, satisfying $\|\mathbf{w}\|_0 = N$ and $\|\mathbf{w}\|_2^2 \leq P_A$, with P_A denoting the maximum transmit power budget. Also, z_l^d is the additive white Gaussian noise component, satisfying $z_l^d \sim \mathcal{CN}(0, \sigma_l^2)$ with σ_l^2 being the received antenna noise power.

²The rationale of FAS resembles traditional antenna selection. From the signal processing viewpoint, they are indeed similar. However, it is worth noting that FAS is designed to switch the antenna's position very finely within a given space, typically handling an enormous number of correlated signals for selection [35], [36].

³The acquisition of Eve's channel information, namely β_{ae} and β_{eb} , can be deduced through the application of scenario-specific prior knowledge or historical statistical data, constituting a general approach for handling unknown Eve channels [37], [38].

⁴To accurately capture the intricate spatial dependencies between antenna ports, our investigation employs a fully correlated channel model [39], which effectively models the mutual coupling and propagation dynamics inherent in reconfigurable antenna arrays. This approach ensures the spatial correlation effects, which are critical for optimizing PLKG.

⁵The investigation primarily focuses on the performance of FAS in generating physical layer keys under passive eavesdropping conditions. Consequently, the impact of pilot contamination is not considered at this stage.

Subsequently, both receivers employ least-squares (LS) estimation techniques [42], [43] to derive channel state information⁶:

$$\hat{h}_b \triangleq s_d^* y_b^d = \mathbf{h}_{ab}^\top \mathbf{w} + \tilde{z}_b^d, \quad (3)$$

$$\hat{h}_e^d \triangleq s_d^* y_e^d = \mathbf{h}_{ae}^\top \mathbf{w} + \tilde{z}_e^d, \quad (4)$$

where $\tilde{z}_b^d = s_d^* z_b^d$ and $\tilde{z}_e^d = s_d^* z_e^d$ represent the noise terms, respectively.

During the uplink stage, Bob transmits the corresponding uplink pilot $s_u \in \mathbb{C}$ with $|s_u|^2 = 1$, yielding received signals at Alice and Eve:

$$\mathbf{y}_l^u = \sqrt{P_B} \mathbf{h}_{lb} s_u + \mathbf{z}_l^u, l \in \{a, e\}, \quad (5)$$

where P_B denotes Bob's transmit power budget. The noise components maintain $\mathbf{z}_l^u \sim \mathcal{CN}(0, \sigma_l^2 \mathbf{I}_M)$.

Through LS estimation, the channel estimates become:

$$\hat{\mathbf{h}}_l^u \triangleq s_u^* \mathbf{y}_l^u = \sqrt{P_B} \mathbf{h}_{lb} + \tilde{\mathbf{z}}_l^u, l \in \{a, e\}, \quad (6)$$

where $\tilde{\mathbf{z}}_l^u = s_u^* \mathbf{z}_l^u$.

To align Alice's vector estimate $\hat{\mathbf{h}}_a^u$ with Bob's scalar estimate $\hat{h}_b$, we construct an equivalent channel measurement at Alice by applying beamforming:

$$\hat{h}_a \triangleq \mathbf{w}^H \hat{\mathbf{h}}_a^u = \sqrt{P_B} \mathbf{w}^H \mathbf{h}_{ab} + z_a^u, \quad (7)$$

where the noise is $z_a^u = \mathbf{w}^H \tilde{\mathbf{z}}_a^u$.⁷

In practice, the constructed channel parameters $\hat{h}_a$ and $\hat{h}_b$ exhibit strong temporal correlation within channel coherence intervals [10], [45], [46]. Following standard PLKG procedures including quantization, information reconciliation, and privacy amplification [10], these correlated measurements are converted into secret keys. During the complete process of a single round of key generation, the FAS port configuration of Alice, such as the combination of N activated ports, remains fixed. Under this condition, the channel reciprocity under the TDD protocol is preserved, since the statistical characteristics of the uplink and downlink channels, including the spatial correlation matrix $\mathbf{J}_A$ and path loss β_{ab} , are fully determined by these fixed port positions. This operational behavior is consistent with that of traditional FA-assisted systems, and therefore no reciprocity degradation is introduced by port switching. In fact, regarding the switching process, the dynamic port switching of FAS occurs only between two adjacent rounds of key generation. After completing the quantization and information reconciliation stages of the previous, the FAS updates its ports configuration to provide new spatial degrees of freedom for the next round of key generation. Importantly, since this reconfiguration takes place outside the channel-probing interval, it does not affect the reciprocity of the completed rounds. Distinct from conventional approaches [45], [46], our contribution emphasizes on optimizing the channel probing phase through joint beamforming design and port selection to maximize the resulting SKR.

⁶The LS is a common technique to obtain estimates in practical systems [44].

⁷The additive noise terms at different nodes during the uplink and downlink channel probing processes all exhibit i.i.d. properties. This model choice is consistent with standard assumptions in physical-layer communication systems, as demonstrated in [37].

III. PROBLEM FORMULATION

In this section, leveraging the channel estimates acquired in Sec. II, we formulate an optimization problem over the beamforming vector $\mathbf{w}$ to improve system performance. To this end, we first derive the SKR given Eve's channel estimates that facilitate its subsequent maximization design through optimization [42]. Specifically, given the observation of Eve's antenna, the SKR is defined as the conditional mutual information of the legitimate parties' channel estimates [45] as follows⁸

$$R_{SK} = \mathcal{I}(\hat{h}_a; \hat{h}_b | \hat{h}_e^d, \hat{h}_e^u). \quad (8)$$

Subsequently, two FAS-assisted PLKG models based on two scenarios were established. In the independent scenario, Eve is assumed to be located at a distance greater than $\frac{\lambda}{2}$ from Bob, such that Eve's channel estimates being independent of those of legitimate users. In the correlated scenario, Eve was considered to be within a distance of $\frac{\lambda}{2}$ from Bob, thereby accounting for the spatial correlation between the eavesdropping channel and the legitimate channel. Approximate SKR expressions are derived for both scenarios by exploiting mutual information and joint entropy analysis, considering both the beamforming and FAS correlation channel model at the BS.

A. Independent and Identically Distributed Scenario

In the i.i.d scenario, Eve's channel is independent of the channels between Alice and Bob. Under this assumption, the SKR is given by [48], [49]

$$R_{SK}^{\text{iid}} = \mathcal{I}(\hat{h}_a; \hat{h}_b) = \log_2 \frac{\mathcal{R}_{aa} \mathcal{R}_{bb}}{\det(\mathbf{R}_{ab})}. \quad (9)$$

The covariance matrix for the channel estimates at Alice and Bob is defined as follows

$$\mathbf{R}_{ab} = \begin{bmatrix} \mathcal{R}_{aa} & \mathcal{R}_{ab} \\ \mathcal{R}_{ba} & \mathcal{R}_{bb} \end{bmatrix}, \quad (10)$$

where $\mathcal{R}_{xy} = \mathbb{E}\{\hat{h}_x \hat{h}_y^H\}$, $x, y \in \{a, b\}$ is the channel covariances of the corresponding channel estimates.

By incorporating the derived channel estimates into the mutual information expression (9) and assuming uniform noise variance ($\sigma_a^2 = \sigma_b^2 = \sigma_e^2 \triangleq \sigma^2$), as commonly adopted in [42], [43], we establish an approximate expression for the SKR through the following lemma.

Lemma 1: The SKR between Alice and Bob is expressed as follows

$$R_{SK}^{\text{iid}} = \log_2 \frac{(P_B \mathbf{w}^H \mathbf{J}_A \mathbf{w} \beta_{ba} + \|\mathbf{w}\|_2^2 \sigma^2) (\mathbf{w}^H \mathbf{J}_A \mathbf{w} \beta_{ba} + \sigma^2)}{(\|\mathbf{w}\|_2^2 + P_B) \sigma^2 \mathbf{w}^H \mathbf{J}_A \mathbf{w} \beta_{ba} + \|\mathbf{w}\|_2^2 \sigma^4}. \quad (11)$$

Proof: Please refer to Appendix A. ■

Thus, the beamforming design could be formulated as

$$\max_{\mathbf{w}} R_{SK}^{\text{iid}} \quad (12a)$$

⁸In this paper, we focus on the investigation of PLKG under the far-field plane-wave assumption for FAS, considering two common eavesdropping presence scenarios [37], [47].

$$\text{s.t. } \|\mathbf{w}\|_2^2 \leq P_A, \quad (12b)$$

$$\|\mathbf{w}\|_0 \leq N, \quad (12c)$$

where constraint (12b) indicates that the transmit beamforming power does not exceed the maximum transmit power budget P_A , while (12c) the number of available RF chains to N .

It could be observed that the optimization problem in (12) is highly nonconvex due to the high-order coupling of optimization variables in the objective function. In addition, the port selection is non-convex combinatorial in nature due to the sparse constraint, which makes problem (12) more challenging to handle.

B. Spatially Correlated Eavesdropping Scenario

In this section, we assume that Eve's downlink channel estimates are correlated with those of legitimate parties. Based on (8), the SKR can be given by

$$R_{SK}^{cc} = \mathcal{I}(\hat{h}_a; \hat{h}_b | \hat{h}_e^d) = \log_2 \frac{\det(\mathbf{R}_{ae} \mathbf{R}_{be})}{\det(\mathbf{R}_{abe}) \mathcal{R}_{ee}}, \quad (13)$$

where the covariance matrices are defined as

$$\mathbf{R}_{ue} = \begin{bmatrix} \mathcal{R}_{uu} & \mathcal{R}_{ue} \\ \mathcal{R}_{eu} & \mathcal{R}_{ee} \end{bmatrix}, u \in \{a, b\}, \quad (14)$$

$$\mathbf{R}_{abe} = \begin{bmatrix} \mathcal{R}_{aa} & \mathcal{R}_{ab} & \mathcal{R}_{ae} \\ \mathcal{R}_{ba} & \mathcal{R}_{bb} & \mathcal{R}_{be} \\ \mathcal{R}_{ea} & \mathcal{R}_{eb} & \mathcal{R}_{ee} \end{bmatrix}. \quad (15)$$

By exploiting Lemma 1, the SKR between Alice and Bob, given the estimate of the channel in Eve, can be given by (16), as shown at the bottom of the page, where $H_u = \beta_{ab} \mathbf{w}^H \mathbf{J}_A \mathbf{w}$, $H_e = \beta_{ae} \mathbf{w}^H \mathbf{J}_A \mathbf{w}$, and $H_{ue} = \sqrt{\beta_{ab} \beta_{ae}} \mathbf{w}^H \mathbf{J}_A \mathbf{w}$.

Remark 1: The analytical expression derived in (16) reveals that the SKR depends solely on the beamforming $\mathbf{w}$ and the channel's statistical information characterized by covariance matrices. Given the quasi-static nature of covariance matrices in dense scattering environments [50], we adopt a practical assumption that these matrices can be effectively estimated from historical channel measurements leveraging well-established estimate techniques [51]. This allows us to concentrate our efforts on developing optimal beamforming and port selection for the FAS-assisted PLKG.

Thus, the beamforming design optimization problem under spatially correlated channel models can be formulated as

$$\max_{\mathbf{w}} R_{SK}^{cc} \quad (17a)$$

$$\text{s.t. (12b), (12c).} \quad (17b)$$

C. Analysis of FAS-Assisted PLKG

The FAS introduces a transformative approach to PLKG by dynamically optimizing antenna port configurations to

exploit spatial correlation effectively. This analysis focuses on maximizing the SKR under transmit power constraints and sparse port activation. The SKR R_{SK} is quantified as the achievable secret bits per channel coherence interval, which can be lower bounded as follows [38], [52]:

$$R_{SK} \geq \mathcal{I}(\hat{h}_a; \hat{h}_b) - \min \left[\mathcal{I}(\hat{h}_a; \hat{h}_e^d), \mathcal{I}(\hat{h}_a; \hat{h}_e^u) \right]. \quad (18)$$

In the i.i.d. scenario, the mutual information between eavesdropping and legitimate nodes is nearly zero [38]. Then, based on (18), we have

$$R_{SK}^{iid} \geq \mathcal{I}(\hat{h}_a; \hat{h}_b). \quad (19)$$

In the spatially correlated eavesdropping scenario, the Eve's downlink estimates are correlated with those of legitimate nodes. Then, we have

$$R_{SK}^{cc} \geq \mathcal{I}(\hat{h}_a; \hat{h}_b) - \mathcal{I}(\hat{h}_a; \hat{h}_e^d). \quad (20)$$

Note that the mutual information between Alice and Bob in (19) and (20), $\mathcal{I}(\hat{h}_a; \hat{h}_b)$, is mainly influenced by $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$. Specifically, the following lemma formally characterizes this dependence and provides guidance for the joint design of $\mathbf{w}$ and FAS port activation.

Lemma 2: The objective function in (12) increases monotonically with $\beta_{ab} \mathbf{w}^H \mathbf{J}_A \mathbf{w}$.

Proof: Please refer to Appendix B. ■

For systems where the number of preset ports M is large, i.e., ($M \gg N$) or the normalized distance between ports is sufficiently small, the spatially correlated matrix at Alice $\mathbf{J}_A$ is dominated by its largest eigenvalues [36]. This property enables approximate optimization based on the largest eigenvalue $\lambda_{\max}$ and its corresponding eigenvector $\mathbf{u}_{\lambda_{\max}}$. As the number of pre-set ports M decreases, the available degrees of freedom for port selection are reduced, thereby exerting a negative influence on the maximum eigenvalue $\lambda_{\max}$. This results in a decrease in the value of $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$, leading to a reduction in the SKR presented in both (11) and (16). In contrast, when M is sufficiently large, an increase in N causes the sparse vector $\mathbf{w}$ to leverage more ports to approximate the eigenvectors $\mathbf{u}_{\lambda_{\max}}$, thereby resulting in an increase in the value of $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$. Consequently, this enhances the SKR in (11) and (16). To determine the beamforming vector $\mathbf{w}$ for the optimal combination of N activated ports, one can exhaustively search all possible port subsets and solve the optimization problem for each, then select the highest-performing configuration. However, this method generally incurs exceedingly high computational complexity. Alternatively, convex relaxation techniques can provide a relatively computationally-efficient sparse beamforming vector $\mathbf{w}$, which will be introduced in next section.

$$R_{SK}^{cc} = \log_2 \left(\frac{(P_B H_u H_e + (P_B H_u + H_e) \|\mathbf{w}\|_2^2 \sigma^2 + \|\mathbf{w}\|_2^2 \sigma^4 - P_B |H_{ue}|^2) (H_u H_e + (H_u + H_e) \sigma^2 + \sigma^4 - |H_{ue}|^2)}{[(P_B + \|\mathbf{w}\|_2^2) (H_u H_e + (H_u + H_e) \sigma^2 + \sigma^4 - |H_{ue}|^2) - P_B \sigma^2 (H_e + \sigma^2)] (H_e + \sigma^2) \sigma^2} \right). \quad (16)$$

IV. JOINT PORT SELECTION AND BEAMFORMING OPTIMIZATION

In this section, we aim to develop an iterative algorithm to address the nonconvex items of the formulated problems via SCA combined with Cauchy-Schwarz inequality relaxation. Then, a reweighted ℓ_1 -norm method is applied to advocate the sparsity in the solution to the port activation problem.

A. Optimization for Problem (12)

Firstly, we optimize the transmit sparse beamforming $\mathbf{w}$ to maximize the SKR $R_{\text{SK}}^{\text{iid}}$ under the i.i.d. scenario. According to Lemma 2, the objective function in problem (12) increases monotonically with $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$. Thus, (12) is equivalent to the following problem

$$\max_{\mathbf{w}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} \quad (21a)$$

$$\text{s.t. (12b), (12c).} \quad (21b)$$

By applying the eigen-decomposition $\mathbf{J}_A = \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H$ and introducing the slack optimization variable Z , problem (21) can be equivalently transformed into

$$\max_{\mathbf{w}, Z} Z \quad (22a)$$

$$\text{s.t. (12b), (12c),} \quad (22b)$$

$$\mathbf{w}^H \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H \mathbf{w} \geq Z. \quad (22c)$$

However, problem (22) is still non-convex due to constraints (12c) and (22c). To tackle (12c), the nonconvex ℓ_0 -norm can be effectively replaced by the sparsity promoting convex ℓ_1 -norm constraint [53], [54], resulting in

$$\|\mathbf{w}\|_1 \leq N. \quad (23)$$

To handle (22c), the iterative SCA technique is applied. In particular, in the l -th iteration of the SCA, a convex subset of (22c) is given by

$$-\mathbf{w}_{l-1}^H \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H \mathbf{w}_{l-1} + 2\text{Re}(\mathbf{w}_{l-1}^H \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H \mathbf{w}) \geq Z, \quad (24)$$

i.e., (24) $\implies$ (22c), where $\mathbf{w}_{l-1}$ is the beamforming vector obtained from the $l-1$ -th iteration of the SCA.

As such, a lower-bound performance to (22) can be obtained by solving

$$\max_{\mathbf{w}, Z} Z \quad (25a)$$

$$\text{s.t. (12b), (23), (24).} \quad (25b)$$

B. Optimization for Problem (17)

To tackle problem (17), we first simplify the optimization problem according to Lemma 2. Without loss of generality, the SKR $R_{\text{SK}}^{\text{cc}}$ in (16) can be defined as $R_{\text{SK}}^{\text{cc}} = \log_2 f(x_0)$, which increases monotonically for x_0 , where

$$x_0 = \beta_{\text{ab}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} - \frac{\|\sqrt{\beta_{\text{ab}} \beta_{\text{ae}}} \mathbf{w}^H \mathbf{J}_A \mathbf{w}\|^2}{\beta_{\text{ae}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} + \sigma^2}. \quad (26)$$

Thus, the original problem (17) is equivalent to the following problem:

$$\max_{\mathbf{w}} \beta_{\text{ab}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} - \frac{\|\sqrt{\beta_{\text{ab}} \beta_{\text{ae}}} \mathbf{w}^H \mathbf{J}_A \mathbf{w}\|^2}{\beta_{\text{ae}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} + \sigma^2} \quad (27a)$$

$$\text{s.t. (12b), (12c).} \quad (27b)$$

To address the nonconvex objective function (27a), slack optimization variables Y , I , and L are introduced to equivalently reformulate problem (27):

$$\max_{\mathbf{w}, Y, I, L} Y - \frac{\|I\|^2}{L} \quad (28a)$$

$$\text{s.t. (12b), (12c),} \quad (28b)$$

$$Y \leq \beta_{\text{ab}} \mathbf{w}^H \mathbf{J}_A \mathbf{w}, \quad (28c)$$

$$L \leq \beta_{\text{ae}} \mathbf{w}^H \mathbf{J}_A \mathbf{w} + \sigma^2, \quad (28d)$$

$$I \geq \sqrt{\beta_{\text{ab}} \beta_{\text{ae}}} \mathbf{w}^H \mathbf{J}_A \mathbf{w}. \quad (28e)$$

By applying the definition $\mathbf{J}_A = \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H$, we have

$$\mathbf{J}_A = \mathbf{U}_A \mathbf{\Lambda}^H \mathbf{U}_A^H = \mathbf{U}_A (\mathbf{\Lambda}^H)^{\frac{1}{2}} (\mathbf{\Lambda}^H)^{\frac{1}{2}} \mathbf{U}_A^H. \quad (29a)$$

Thus, problem (28) can be rewritten by

$$\max_{\mathbf{w}, Y, I, L} Y - \frac{\|I\|^2}{L} \quad (30a)$$

$$\text{s.t. (12b), (12c),} \quad (30b)$$

$$Y \leq \beta_{\text{ab}} \mathbf{w}^H \mathbf{U}_A (\mathbf{\Lambda}^H)^{\frac{1}{2}} (\mathbf{\Lambda}^H)^{\frac{1}{2}} \mathbf{U}_A^H \mathbf{w}, \quad (30c)$$

$$L \leq \beta_{\text{ae}} \mathbf{w}^H \mathbf{U}_A (\mathbf{\Lambda}^H)^{\frac{1}{2}} (\mathbf{\Lambda}^H)^{\frac{1}{2}} \mathbf{U}_A^H \mathbf{w} + \sigma^2, \quad (30d)$$

$$I \geq \beta_{\text{abe}} \mathbf{w}^H \mathbf{U}_A (\mathbf{\Lambda}^H)^{\frac{1}{2}} (\mathbf{\Lambda}^H)^{\frac{1}{2}} \mathbf{U}_A^H \mathbf{w}, \quad (30e)$$

where $\beta_{\text{abe}} = \sqrt{\beta_{\text{ab}} \beta_{\text{ae}}}$.

Note that the objective function is now jointly concave with respect to the optimization variables. However, constraints (30c) and (30d) are both nonconvex due to right-hand-side (RHS) term. To handle the constraints, similar to the previous section, based on the SCA technique, we provide the following lemma to establish a performance lower bound.

Lemma 3: According to Cauchy-Schwarz inequality $\mathbf{x} \mathbf{y}^H \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$, the inequality during l -th iteration can be given by

$$\|\mathbf{f}\|_2 \geq \frac{\text{Re}(\mathbf{f} \mathbf{f}_{l-1}^H)}{\|\mathbf{f}_{l-1}^H\|_2}. \quad (31)$$

Proof: Please refer to Appendix C. ■

By defining $\mathbf{f} = \mathbf{w}^H \mathbf{U}_A (\mathbf{\Lambda}^H)^{\frac{1}{2}}$, and applying the sparsity-promoting convex ℓ_1 -norm [53] to approximate the nonconvex constraints (12c) in (30), a suboptimal solution to (30) can be obtained via solving the following optimization problem:

$$\max_{\mathbf{w}, Y, I, L} Y - \frac{\|I\|^2}{L} \quad (32a)$$

$$\text{s.t. } \|\mathbf{w}\|_2^2 \leq P_A, \quad (32b)$$

$$\|\mathbf{w}\|_1 \leq N, \quad (32c)$$

$$Y \leq \frac{\beta_{\text{ab}} \text{Re}(\mathbf{f} \mathbf{f}_{l-1}^H)}{\|\mathbf{f}_{l-1}^H\|_2}, \quad (32d)$$

$$L \leq \frac{\beta_{\text{ae}} \text{Re}(\mathbf{f} \mathbf{f}_{l-1}^H)}{\|\mathbf{f}_{l-1}^H\|_2} + \sigma^2, \quad (32e)$$

$$I \geq \beta_{\text{abe}} \mathbf{f} \mathbf{f}^H. \quad (32f)$$

Algorithm 1 Iterative Algorithm for Solving (12) or (17)**Require:** Threshold ε_0 and covariance matrix $\mathbf{J}_A$.

- 1: Set iteration index: $l = 0$.
- 2: Initial: $\mathbf{w}^{(0)}$.
- 3: **repeat**
- 4: For solving (12):
- 5: Update $\mathbf{w}^{(l+1)}$ by solving Problem (25).
- 6: Calculate the objective value $R^{(l)}$ of Problem (11).
- 7: For solving (17):
- 8: Update $\mathbf{w}^{(l+1)}$ by solving Problem (32).
- 9: Calculate the objective value $R^{(l)}$ of Problem (16).
- 10: $l \leftarrow l + 1$.
- 11: **until** $|R^{(l)} - R^{(l-1)}| \leq \varepsilon_0$.

C. Overall Algorithm

An iterative algorithm is provided to obtain an approximate solution for (12) and (17), respectively.

To attain the required number of iterations $\|\mathbf{w}\|_0 = N$, we develop an iterative reweighted ℓ_1 -norm approach for constraints (23) and (32c), introducing a diagonal weighting matrix $\mathbf{V}$ [53]. Especially, at the l -th iteration, it is denoted as $\mathbf{V}_l = g(\mathbf{w}_{l-1})$. To promote sparsity, $g(\mathbf{w}_{l-1})$ can be given by

$$g(\mathbf{w}_{l-1}) \triangleq \text{diag} \left(\frac{1}{|[\mathbf{w}_{l-1}]_1| + \gamma}, \dots, \frac{1}{|[\mathbf{w}_{l-1}]_M| + \gamma} \right), \quad (33)$$

where $[\mathbf{w}_{l-1}]_m$ denotes the m -th element of $\mathbf{w}$ at the $(l-1)$ -th iteration and γ is a small regularization parameter to avoid division by zero.⁹

Remark 2: The regularization parameter γ is crucial for regulating the sparsity of the beamforming vector. At the l -th iteration, the diagonal elements of the weighting matrix $\mathbf{V}_l$ are related to the elements $[\mathbf{w}_{l-1}]_m$. This adaptive scaling maintains numerical stability by stopping weight vector components from growing uncontrollably as $[\mathbf{w}_{l-1}]_m$ nears zero. The parameter γ prevents the iterative optimization from diverging when the components of $\mathbf{w}$ approach the null space [55].

Based on constraints (23) and (32c), the weighted ℓ_1 constraint can be expressed as

$$\|\mathbf{V}_l \mathbf{w}\|_1 \leq N. \quad (34)$$

The reweighted ℓ_1 norm iterative algorithm is summarized in Algorithm 2.

D. Complexity Analysis

The computational complexity of the proposed reweighted algorithm Algorithm 2 is $\mathcal{O}(M^3 + M^2N + M)$, which can be attributed to three principal components, including eigen-decomposition with a complexity of $\mathcal{O}(M^3)$, SCA iterations with a complexity of $\mathcal{O}(M^2N)$, and the reweighting computation with a complexity of $\mathcal{O}(M)$.

⁹Combined with the domain consensus on reweighted ℓ_1 -norm optimization [53] and the characteristics of the system parameters in this paper, the value of γ is 10^{-3} .

Algorithm 2 The Reweighted Algorithm for Solving (12) or (17)**Require:** Threshold ε_0 and covariance matrix $\mathbf{J}_A$.

- 1: Set: $r = 0$.
- 2: Initial: $\mathbf{w}^{(0)}$ and $\mathbf{V} = \mathbf{I}_M$.
- 3: **repeat**
- 4: Solve problem (12) or (17) with a weighted ℓ_1 constraint adopting **Algorithm 1**.
- 5: Calculate the objective value $\mathbf{V}^{(r+1)} = g(\mathbf{w}^{(r)})$.
- 6: $r \leftarrow r + 1$.
- 7: **until** $|R^r - R^{(r-1)}| \leq \varepsilon_0$.

V. SLIDING WINDOW-BASED PORT SELECTION

In Section IV, a convex relaxation technique was employed to derive the computationally efficient sparse beamforming vector $\mathbf{w}$. Alternatively, it is also possible to analyze the FAS-assisted PLKG and develop a low-complexity heuristic algorithm that serves as a practical substitute for exhaustive searching methods, which typically exhibit high computational complexity. According to Lemma 2, the mutual information between Alice and Bob, $\mathcal{I}(\hat{h}_a; \hat{h}_b)$, in (19) and (20), is influenced by the largest eigenvalue, $\lambda_{\max}$. In this section, leveraging Rayleigh-quotient theory [56], we provide the analysis of the largest eigenvalue, which is influenced by zero-order Bessel function of the first kind. Based on the analysis, a sliding window-based port selection is proposed to obtain the activated port selection for FAS-assisted PLKG. Firstly, the following discussion about properties of the zero-order Bessel function of the first kind is conducted:

Remark 3: The zero-order Bessel function of the first kind expressed as: $J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k}$, is an oscillatory decaying function within the interval $[-1, 1]$ as x increases, with gradually decreasing amplitude. This function reaches its maximum value $J_0(0) = 1$ at $x = 0$, and then fluctuates between positive and negative values and approaches 0 as x increases [40].

The (n, m) -th element of matrix $\mathbf{J}_A$ is given by [36]:

$$\mathbf{J}_A(n, m) = J_0 \left(\frac{2\pi|n-m|}{M-1} W \right). \quad (35)$$

As W increases, the argument $\frac{2\pi|n-m|}{M-1} W$ increases for $\forall n \neq m$. Due to the oscillatory decaying property of $J_0(x)$ based on Remark 3, the sum of absolute values of the non-diagonal elements of matrix $\mathbf{J}_A$ gradually decreases.

Based on the Rayleigh-quotient theory, the eigenvalues of the symmetric matrix $\mathbf{J}_A$ satisfy the following properties:

Remark 4: For a non-zero M -dimensional vector $\mathbf{x}$, the Rayleigh-quotient of matrix $\mathbf{J}_A \in \mathbb{C}^{M \times M}$, is defined as: $R(\mathbf{x}) = \frac{\mathbf{x}^H \mathbf{J}_A \mathbf{x}}{\mathbf{x}^H \mathbf{x}}$. The maximum eigenvalue $\lambda_{\max}$ and minimum eigenvalue $\lambda_{\min}$ of matrix $\mathbf{J}_A$ satisfy: $\lambda_{\min} \leq R(\mathbf{x}) \leq \lambda_{\max}$.

As W increases, the sum of absolute values of the non-diagonal elements of matrix $\mathbf{J}_A$ decreases. From the perspective of the Rayleigh-quotient, for any vector $\mathbf{x}$, we have:

$$\mathbf{x}^H \mathbf{J}_A \mathbf{x} = \sum_{n=1}^M \sum_{m=1}^M x_n J_{n,m} x_m. \quad (36)$$

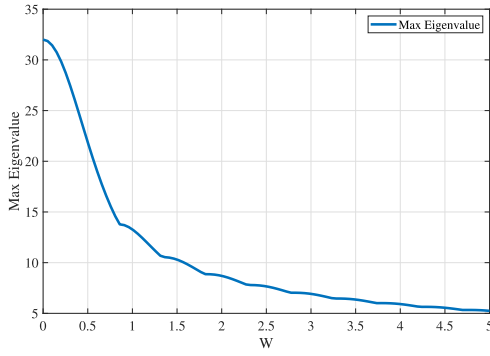


Fig. 2. The maximum eigenvalue of matrix J_A versus W .

The diagonal elements of matrix J_A are $J_{n,n} = J_0(0) = 1$, which are independent of W . The decrease in magnitude of the non-diagonal elements leads to a relatively smaller value of $\mathbf{x}^H J_A \mathbf{x}$, thereby causing the maximum eigenvalue to decline, as shown in Fig. 2.

Lemma 2 suggests that a decrease in $\lambda_{\max}$ directly reduces $\mathbf{w}^H J_A \mathbf{w}$ and consequently lowers the SKR of FAS-assisted communications. To counteract this effect by exploiting closely spaced active ports, we propose a sliding-window initialization over the one-dimensional array. The key steps of this method are summarized in the following:

Step 1: Candidate Window Sets Generation. Generate $M-N+1$ candidate window sets by sliding a window S of length N across the M ports.¹⁰

Step 2: Sliding Window Selection. Compute $\mathbf{w}^H J_A \mathbf{w}$ for each window set and select the window set with the maximum value to serve as the initial beamforming vector $\mathbf{w}^{(0)}$.

Step 3: Power Optimization. Perform power optimization on the beamforming vector $\mathbf{w}^{(0)}$ within the selected window set to satisfy the transmit power constraint.

The proposed window selection process efficiently reduces the initialization overhead while leveraging the spatial correlation property of the FAS. Especially, compared to exhaustively traversing all possible port subset combinations (e.g., the number of combinations is more than 200,000 for $M = 32, N = 5$), the computational complexity is reduced significantly from exponential to polynomial time $\mathcal{O}((M+N-1)M^2)$.¹¹ The sparse vector and transformation matrix computation of the sliding window selection-based algorithm is summarized in Algorithm 3.

Defining $\bar{\mathbf{w}} = S\mathbf{w}$, the sliding window selection-based iterative algorithm can be given by Algorithm 4.

VI. SIMULATION RESULTS

In this section, we provide simulation results to illustrate the PLKG performance of the proposed method and investigate the impact of spatial correlation on the SKR.

¹⁰According to Remark 3, while the window length is smaller than N , the activated ports must be aggregated across multiple discrete windows, which lead to significant attenuation of the $J_0(\cdot)$ values between ports and ultimately causes a reduction in the SKR.

¹¹The algorithm can be extended by considering the worst-case selection problem with K_e multiple Eves. In terms of the adaptability of port selection strategies, the objective function is modified to maximize the SKR of the legitimate channel and minimize the maximum information leakage of the high-threat Eve. Then, worst-case optimization is executed on the Eve, reducing the complexity from $\mathcal{O}(K_e(M+N-1)M^2)$ to $\mathcal{O}(M+N-1)M^2$.

Algorithm 3 Sparse Vector and Transformation Matrix Computation

Require: symmetric matrix $J_A \in \mathbb{R}^{M \times M}$, selection dimension N , Power budget P_A .

- 1: Compute eigenvalues and eigenvectors: $[V, D]$.
- 2: $\lambda_{\max} = \max(\text{diag}(D))$, index for maximum value $I = \text{argmax}(\text{diag}(D))$.
- 3: Extract the eigenvector corresponding to the maximum eigenvalue: $\mathbf{u} = V(I)$.
- 4: Initialize $\mathbf{w}$, maximum value $F_{\max}$.
- 5: **for** $s = 1$ to $M-N+1$ **do**
- 6: Determine the index set $\mathcal{I}$ covered by the current sliding window: $\mathcal{I} \leftarrow [s, s+1, \dots, s+N-1]$.
- 7: Initialize the current sparse vector $\mathbf{w}$ as a zero vector: $\mathbf{w}^{\text{ini}} = \mathbf{0}_M$.
- 8: Assign the values of $\mathbf{u}$ at the corresponding indices to $\mathbf{w}$: $\mathbf{w}(\mathcal{I}) = \mathbf{u}(\mathcal{I})$.
- 9: Calculate F_{cur}
- 10: **if** $F_{\text{cur}} > F_{\max}$ **then**
- 11: Update: $F_{\max} = F_{\text{cur}}$, $\mathbf{w}^{\text{ini}} = \mathbf{w}$, $\mathcal{I}^* = \mathcal{I}$,
- 12: **end if**
- 13: **end for**
- 14: Normalized $\mathbf{w}^{\text{ini}}$ with power budget P_A .
- 15: Construct the transformation matrix: $S = \mathbf{0}_{M \times M}$, $S(\mathbf{i}^{\text{ini}}, \mathbf{i}^{\text{ini}}) = \mathbf{I}_N$.
- 16: **Output:** $\mathbf{w}^{\text{ini}}$, S .

Algorithm 4 The Sliding Window Selection-Based Algorithm

Require: Threshold ε_0 , covariance matrix J_A , the number of activated port N and power budget P_A .

- 1: Set: $r = 0$.
- 2: Initial: Use **Algorithm 3** to obtain initial beamforming vector $\mathbf{w}^{\text{ini}}$ and sliding window S .
- 3: **repeat**
- 4: For solving (12):
- 5: Update $\bar{\mathbf{w}}^{(l+1)}$ by solving Problem (25).
- 6: Calculate the objective value $R^{(l)}$ of Problem (11).
- 7: For solving (17):
- 8: Update $\bar{\mathbf{w}}^{(l+1)}$ by solving Problem (32).
- 9: Calculate the objective value $R^{(l)}$ of Problem (16).
- 10: $l \leftarrow l + 1$.
- 11: **until** $|R^{(l)} - R^{(l-1)}| \leq \varepsilon_0$.

A. Simulation Settings

Specific parameters are detailed in Table I. In the simulation, Alice and Bob are positioned at coordinates (0 m, 0 m) and (70 m, 0 m), respectively. For the spatially correlated eavesdropping scenario, Eve's position is uniformly distributed within a circular region centered at Bob and radius $r = \lambda/2$ m. In contrast, for the i.i.d. scenario, Eve's location is randomly distributed in distinct regions to ensure statistical independence of eavesdropping channels such as (35 m, 35 m).

The considered algorithms are defined as follows:

- **“FA Opt”**: In this case, Alice is equipped with conventional fixed antennas, and the beamforming is optimized with the iterative algorithm **Algorithm 1**.

TABLE I
PARAMETER TABLE

Parameter	Meaning	Value
M	The number of FAS pre-set port	32 [36]
W	Normalized distance of FAS	0.5λ [36]
N	The number of FAS active antenna	5 [36]
σ^2	Noise variance	-80 dBm [37]
P_A	Power budget of Alice	20 dBm [37]
P_B	Power budget of Bob	20 dBm [37]
γ_0	Path loss at 1 m	-30 dB [37]
α_0	Path loss exponent	2 [37]
ϵ	Convergence threshold	10^{-4} [37]

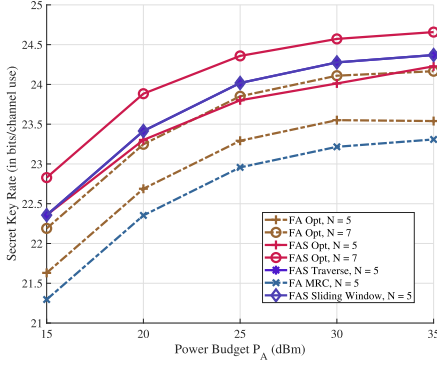


Fig. 3. Secret key rate R_{SK}^{iid} versus power budget P_A in i.i.d. channel with different schemes between FA and FAS.

- **“FA MRC”**: In this case, Alice is equipped with conventional fixed antennas, and the beamforming is optimized with the MRC technique as in [19], [22], [28], and [34].
- **“FAS Opt”**: In this case, Alice is equipped with an FAS, and the beamforming is optimized with the iterative algorithm **Algorithm 1** while activated ports are given by the reweighted algorithm **Algorithm 2**.
- **“FAS Traverse”**: In this case, Alice is equipped with FAS, and the activated port combination is obtained by traversing all possible combinations [57].
- **“FAS Sliding Window”**: In this case, Alice is equipped with FAS. The optimal beamforming and activated ports are obtained by using **Algorithm 4**.
- **“FAS Sliding Window/no”**: In this case, Alice is equipped with FAS, and only activated ports are obtained by exploiting **Algorithm 3**.

B. Impact of the Power Budget

As demonstrated in Fig. 3 and Fig. 4, the SKR of all schemes exhibits a monotonically increasing trend with the transmit power budget P_A in both independent and spatially correlated channel scenarios. This behavior aligns with the theoretical analysis presented in Lemma 1 and Lemma 2, which link the SKR to mutual information that depends on $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ and transmit power budget. Notably, as P_A increases, the SKR growth gradually slows and approaches saturation. This saturation is theoretically underpinned by Lemma 4, which proves that the derivative of the SKR with respect to P_A diminishes asymptotically to zero as P_A becomes sufficiently

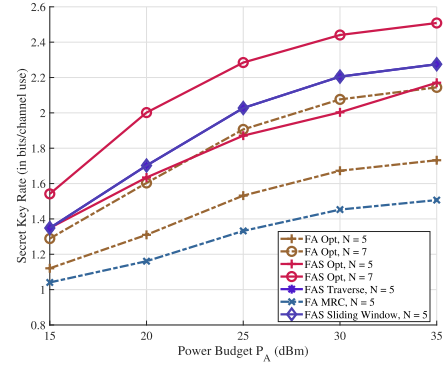


Fig. 4. Secret key rate R_{SK}^{cc} versus power budget P_A in spatially correlated channel with different schemes between FA and FAS.

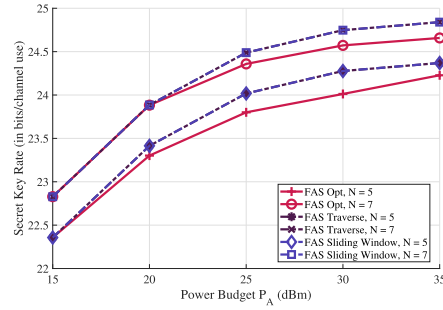


Fig. 5. Secret key rate R_{SK}^{iid} versus power budget P_A in i.i.d. channel with different schemes by exploiting FAS.

large. Therefore, unilaterally increasing Alice’s power budget P_A to a sufficiently large extent cannot lead to significant improvement in the SKR, but instead increases the energy consumption of hardware devices.

Comparing the SKR between FAS and FA, it can be observed that the FAS-based schemes, including “FAS Opt”, “FAS Traverse”, and “FAS Sliding Window”, have significantly higher SKR than the FA-based schemes. It is noteworthy that the FAS scheme achieves similar or better performance with fewer activated antennas (“FAS Opt, $N = 5$ ”) than the FA scheme with more activated antennas (“FA Opt, $N = 7$ ”). This superiority stems from dynamic sparse port selection, which concentrates transmit power on spatially correlated ports to maximize $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$, thereby enhancing the mutual information more efficiently than a fixed-antenna arrangement.

Furthermore, the comparisons of the SKR between different FAS-based schemes are shown in Fig. 5 and Fig. 6. There is a performance gap between the “FAS Opt” scheme based on the reweighted ℓ_1 -norm optimization algorithm and “FAS Traverse” by traversing all possible port combinations. This gap arises because the regularization parameter in (33) causes the direction of port selection to deviate from the optimal port combination with the increase of the power budget. In contrast, the “FAS Sliding Window” effectively guides the optimization process to approach the optimal port combination by selecting consecutive highly correlated ports as the initial values based on the Rayleigh-quotient analysis and the spatial correlation property of the Bessel function. As a result, it consistently

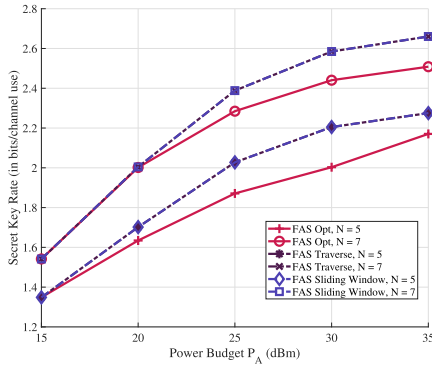


Fig. 6. Secret key rate R_{SK}^{cc} versus power budget P_A in spatially correlated channel with different schemes by exploiting FAS.

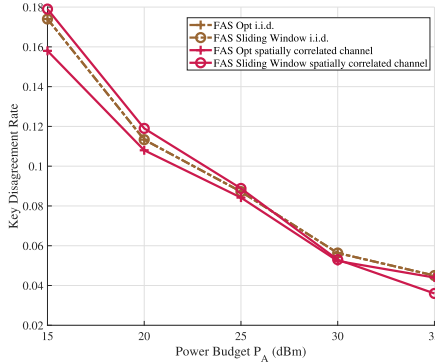


Fig. 7. Key disagreement rate versus power budget P_A with different schemes by exploiting FAS.

achieves performance levels similar to those of an exhaustive search, but with substantially lower complexity.

As demonstrated in Fig. 7, we present the key disagreement rate (KDR) versus transmit power for both scenarios with FAS schemes. It can be observed that “FAS Sliding Window” achieves the KDR comparable to “FAS Opt” with lower computational complexity. The KDR of all schemes decreases with transmit power, as high SNR reduces disagreements between Alice and Bob. According to Remark 4, the value of quadratic form $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ increases with $\|\mathbf{w}\|_2^2$, which is directly influenced by the power budget P_A . Consequently, a larger power budget serves to enhance $\|\mathbf{w}\|_2^2$ and improve the SNR, thereby reducing disagreements between Alice and Bob.

C. Impact of RF Chains

Fig. 8 and Fig. 9 illustrate that the SKR increases monotonically with the number of activated antennas N in both channel scenarios, a direct consequence of extended spatial degrees of freedom. As N grows, the sparse beamforming vector can more accurately approximate the eigenvector corresponding to the largest eigenvalue $\lambda_{\max}(\mathbf{J}_A)$, as discussed in Lemma 2 and Rayleigh-quotient analysis. Indeed, for the term $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ in (11) and (16), an increase in the number of activated ports within fixed available pre-set ports yields a corresponding augmentation in the available spatial degrees of freedom for the beamforming vector $\mathbf{w}$. Consequently, this facilitates a more efficient implementation of channel spatial correlation,

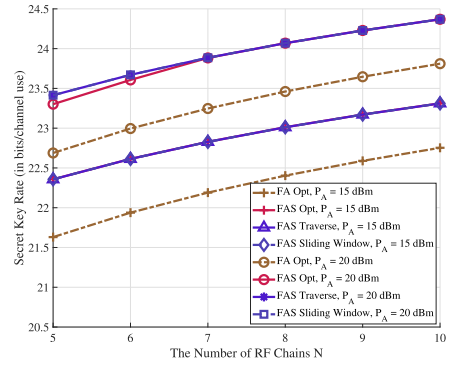


Fig. 8. Secret key rate R_{SK}^{iid} versus the number of RF chains N in i.i.d. channel with different power budget P_A .

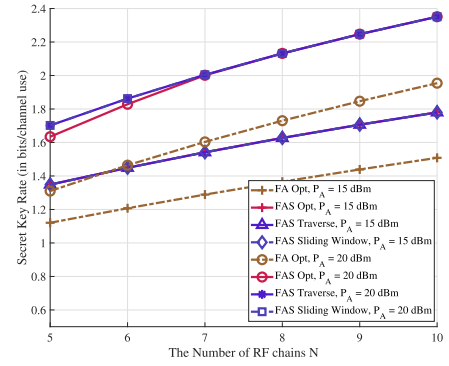


Fig. 9. Secret key rate R_{SK}^{cc} versus the number of RF chains N in spatially correlated channel with different power budget P_A .

which aligns with the predominant eigenmode of the spatial correlation matrix $\mathbf{J}_A$, subsequently maximizing the mutual information between legitimate channels.

Notably, comparing the growth rates of SKR with respect to N in both scenarios, as demonstrated in Fig. 8 and Fig. 9, reveals that SKR increases by 38.2% in the spatially correlated scenario and 4.2% in the i.i.d. scenario, indicating a significantly higher sensitivity of SKR to the number of RF chains in the former. This disparity arises because the SKR of the i.i.d. scenario, R_{SK}^{iid} , is independent of Eve’s channel and its growth depends only on $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$, whereas the SKR of the spatially correlated scenario, R_{SK}^{cc} , relies on conditional mutual information that accounts for Eve’s channel. Thus, activating more ports improves the SKR by both strengthening the legitimate mutual information of the channel via increasing $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ and mitigating the part related to Eve by reducing $\frac{|\sqrt{\beta_{ab}\beta_{ac}}\mathbf{w}^H \mathbf{J}_A \mathbf{w}|^2}{\beta_{ac}\mathbf{w}^H \mathbf{J}_A \mathbf{w} + \sigma^2}$.

Furthermore, as demonstrated in Fig. 8 and Fig. 9, a discrepancy in performance emerges between the “FAS Opt” and “FAS Traverse” methods for $N < 7$. This discrepancy arises because the initial random port activation and regularization parameter in (33) of the reweighted ℓ_1 -norm algorithm deviates from the optimal port combination with fewer activated ports. In contrast, the “FAS Sliding Window” scheme initially selects highly correlated consecutive ports by leveraging the window function $\mathbf{S}$ and optimizes the beamforming vector $\mathbf{w}$ without the regularization parameter, thus improving the SKR.

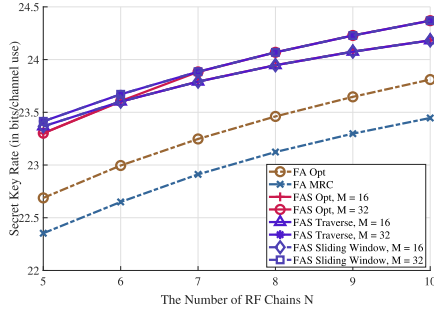


Fig. 10. Secret key rate R_{SK}^{iid} versus the number of RF chains N in i.i.d. channel with different pre-set ports M .

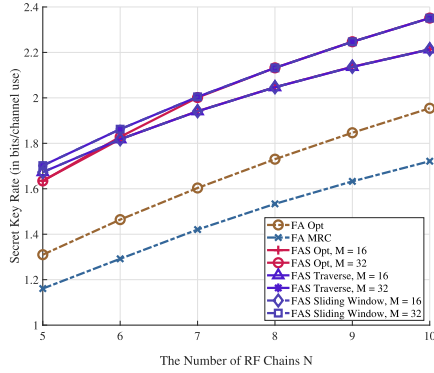


Fig. 11. Secret key rate R_{SK}^{cc} versus the number of RF chains N in spatially correlated channel with different pre-set ports M .

In addition, the number of pre-set ports M also influences the SKR of FAS as shown in Fig. 10 and Fig. 11. This highlights that increasing the number of activated antennas enhances the mutual information of the legitimate channel by extending the spatial degrees of freedom. The FAS-based schemes exhibit consistently higher SKR than the FA-based schemes. In fact, the decrease of the number of FAS preset ports M limits the exploitation of spatial correlation by FAS. When $M = 16$, the SKR of “FAS Opt” for $N = 10$ in the independent channel decreases compared to $M = 32$. This occurs because a smaller M shrinks the available subset of highly correlated ports, limiting the potential to maximize the mutual information of the channel through the term $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ in (11). Therefore, when M is not sufficiently large, the SKR is limited by the number of preset ports M , which aligns with the analysis in Section III-C.

D. Impact of Normalized Size

In Fig. 12 and Fig. 13, as the normalized size W of the FAS increases, the SKR decreases monotonically, primarily due to the reduced spatial correlation between activated ports. The reason is that the SKR is monotonically increasing with $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$, which is directly influenced by $\lambda_{\max}(\mathbf{J}_A)$ since $\mathbf{w}^H \mathbf{J}_A \mathbf{w} \leq \lambda_{\max}(\mathbf{J}_A) \|\mathbf{w}\|_2^2$ according to Rayleigh-quotient theory. From both physical and theoretical perspective, dynamic port selection efficiently performs matching to the spatial correlation characteristics of the channel. When the FAS selects consecutive ports with small spacing, the corresponding Bessel function values $J_0(\cdot)$ between ports

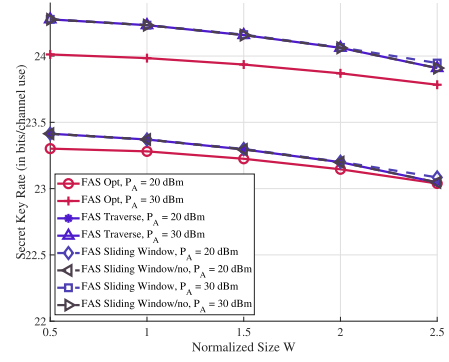


Fig. 12. Secret key rate R_{SK}^{iid} versus normalized size of FAS W in i.i.d. channel with different schemes.

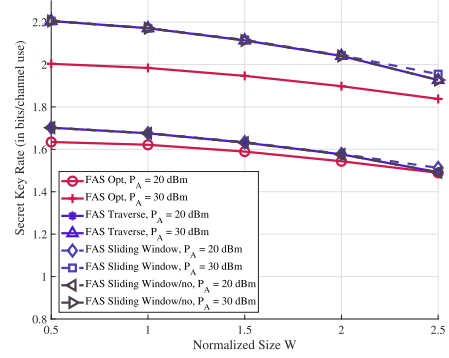


Fig. 13. Secret key rate R_{SK}^{cc} versus normalized size of FAS W in spatially correlated channel with different schemes.

remain close to unit, as analyzed in Remark 3. As a result, the off-diagonal elements of $\mathbf{J}_A$ decay slowly, and $\lambda_{\max}(\mathbf{J}_A)$ is maintained at a high level. In contrast, if ports with large spacing are selected, the oscillatory decay of $J_0(\cdot)$ leads to a reduction in the absolute values of the off-diagonal elements of $\mathbf{J}_A$, causing $\lambda_{\max}(\mathbf{J}_A)$ to decrease. Consequently, $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$ decreases accordingly, and both the mutual information and SKR decline as well. Thus, a smaller $\lambda_{\max}(\mathbf{J}_A)$ under larger W reduces $\mathbf{w}^H \mathbf{J}_A \mathbf{w}$, leading to diminished mutual information $\mathcal{I}(\hat{h}_a; \hat{h}_b)$ and lower SKR. It can be observed that “FAS Sliding Window” maintains a higher SKR than “FAS Opt” across different W , demonstrating that sliding window-based port selection can determine effective activated port combination such that the SKR is improved, as guided by the Bessel function’s decay characteristics and the Rayleigh-quotient theory, as analyzed in Remark 4. It can be noticed that when W becomes large such as $W = 2.5$, the combination of highly correlated ports and beam shaping optimization “FAS Sliding Window” can further increase the SKR compared to simply considering high correlation ports “FAS Sliding Window/no”, because the increase of W results in a decrease in the maximum eigenvalue of $\mathbf{J}_A$, and the resulting sparse beam shaping vector $\mathbf{w}$ based on the corresponding eigenvector $\mathbf{u}_{\max}$, requires further optimization to satisfy the optimal power allocation.

E. Convergence

Fig. 14 and Fig. 15 depict the achieved SKR versus the number of the reweighted ℓ_1 -norm method iterations in both

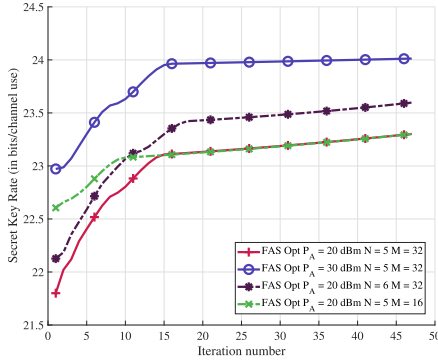


Fig. 14. The convergence of the reweighted ℓ_1 -norm method in i.i.d. channel.

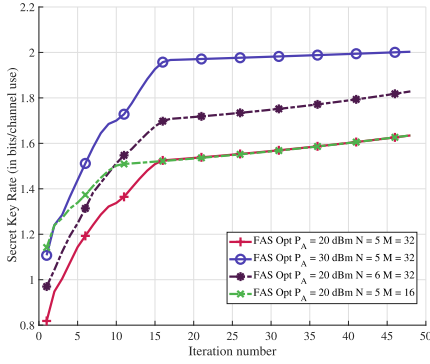


Fig. 15. The convergence of the reweighted ℓ_1 -norm method in spatially correlated channel.

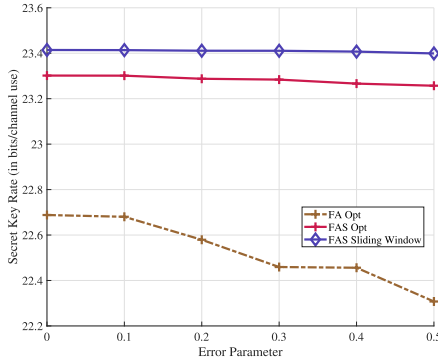


Fig. 16. The robustness of the proposed method in i.i.d. channel.

channel scenarios. As observed, the SKR converges rapidly to stable values after a few iterations on average. For $M = 32$, the proposed algorithm converges after approximately 15 iterations on average. For the case with few FAS pre-set port M , i.e., $M = 16$, the number of iterations required for the convergence is 10 on average. This is because the solution space shrinks when fewer optimization variables are involved and thus fewer iterations are required for convergence.

F. Robustness of Spatial Correlation Matrix

To validate the impact of spatial correlation matrix estimation errors on the proposed algorithm, the estimation error is modeled as $\Delta \mathbf{J}$ in $\mathbb{C}^{M \times M}$, which is assumed to be a symmetric perturbation matrix [39]. The elements $\Delta J(n, m)$ follow a zero-mean complex Gaussian distribution $\mathcal{CN}(0, \sigma_{\Delta}^2)$,

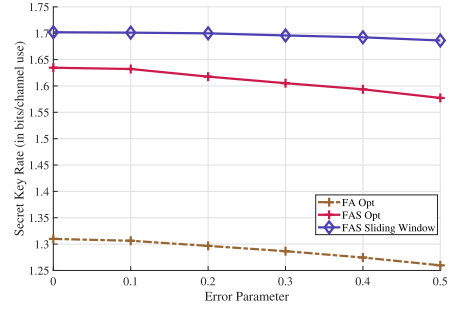


Fig. 17. The robustness of the proposed method in spatially correlated channel.

where σ_{Δ}^2 denotes the error parameter. The case $\sigma_{\Delta}^2 = 0$ corresponds to the ideal error-free scenario, while larger values of σ_{Δ}^2 represent a more significant error. The simulation results presented in Fig. 16 and Fig. 17 provide a quantitative validation of the robustness of the proposed schemes against estimation errors. As the estimation error increases, the achieved SKR decreases for all considered approaches. However, the proposed FAS-based algorithm consistently outperforms the FA-based algorithm and exhibits a slower degradation trend. This assertion is rooted in the findings outlined in Lemma 2, which establishes that the SKR is closely related to the maximum eigen-value of the spatial correlation matrix. It is evident that the number of predetermined ports M in FAS-assisted PLKG substantially exceeds the traditional FA-assisted PLKG with N ports, thereby demonstrating a greater degree of robustness.

VII. CONCLUSION

This paper investigated PLKG in multi-antenna base station systems by applying the FAS to dynamically optimize radio environments. We proposed novel FAS-assisted PLKG models that jointly integrate transmit beamforming and sparse port selection under both i.i.d. and spatially correlated channels, and derived approximate SKR expressions via reciprocal channel probing while explicitly considering eavesdropper channels in correlated environments. The resulted nonconvex optimization problem for maximizing SKR under power and sparse port constraints was addressed by exploiting SCA technique, Cauchy-Schwarz inequality, and a reweighted ℓ_1 -norm algorithm. To improve optimization initialization, a sliding window-based port selection method was introduced utilizing Rayleigh-quotient theory. Simulations demonstrated that the FAS-assisted PLKG scheme significantly outperforms traditional FA-assisted PLKG in both channel environments. Finally, the sliding window approach proposed in this paper approximated the optimal port selection with the lower complexity compared to the reweighted ℓ_1 -norm method and the traverse method, offering valuable insights for practical systems.

APPENDIX

A. Proof of Lemma 1

First, we derive the covariance of channel estimate $\hat{h}_a$ as

$$\mathcal{R}_{aa} = P_B \mathbf{w}^H \mathbb{E} \{ \mathbf{h}_{ab} \mathbf{h}_{ab}^H \} \mathbf{w} + \|\mathbf{w}\|_2^2 \sigma_a^2. \quad (37)$$

By exploiting the channel correlation matrix, the first term of the right-hand side of (37) is calculated as $\mathbb{E}\{\mathbf{h}_{ab}\mathbf{h}_{ab}^H\} = \beta_{ab}\mathbf{J}_A$.

Similarly, we have

$$\mathcal{R}_{bb} = \beta_{ab}\mathbf{w}^H\mathbf{J}_A\mathbf{w} + \sigma_b^2, \quad (38)$$

$$\mathcal{R}_{ab} = \mathcal{R}_{ba} = \sqrt{P_B}\beta_{ab}\mathbf{w}^H\mathbf{J}_A\mathbf{w} = \mathcal{R}_{ba}^H, \quad (39)$$

$$\mathcal{R}_{ee} = \beta_{ae}\mathbf{w}^H\mathbf{J}_A\mathbf{w} + \sigma_e^2, \quad (40)$$

$$\mathcal{R}_{ae} = \sqrt{P_B}\sqrt{\beta_{ab}}\sqrt{\beta_{ae}}\mathbf{w}^H\mathbf{J}_A\mathbf{w} = \mathcal{R}_{ea}^H, \quad (41)$$

$$\mathcal{R}_{be} = \sqrt{\beta_{ab}}\sqrt{\beta_{ae}}\mathbf{w}^H\mathbf{J}_A\mathbf{w} = \mathcal{R}_{eb}^H. \quad (42)$$

Then, we can calculate the two determinants in (9) and obtain the SKR as (1).

B. Proof of Lemma 2

First, we denote $\mathbf{w} = \sqrt{P}\mathbf{w}_0$ with $\|\mathbf{w}_0\|_2^2 = 1$. Since $\frac{dR_{SK}}{dP} > 0$, the optimal P is achieved when $P = P_A$. Then, consider the function

$$f(x) = \frac{(P_Bx + P_A\sigma^2)(x + \sigma^2)}{(P_A\sigma^2 + P_B\sigma^2)x + P_A\sigma^4}, \quad (43)$$

that is monotonically increasing for $x > 0$, since

$$\frac{df(x)}{dx} = \frac{P_B}{\sigma^2} \frac{(P_A + P_B)x^2 + 2P_A\sigma^2x}{((P_A + P_B)x + P_A\sigma^2)^2} > 0. \quad (44)$$

Denote $x = \beta_{ba}\mathbf{w}^H\mathbf{J}_A\mathbf{w}$ and the objective function is $R_{SK} = \log_2 f(x)$. This completes the proof.

C. Proof of Lemma 3

For two complex numbers, $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, we have

$$\begin{aligned} \text{Re}(z_1 z_2^*) &= a_1 a_2 + b_1 b_2 \\ &\leq \sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}. \end{aligned} \quad (45)$$

Letting $z_1 = f$ and $z_2 = f_{l-1}$ yields the desired inequality.

D. Proof of Lemma 4

Denote $\mathbf{w} = \sqrt{P}\mathbf{w}_0$ with $\|\mathbf{w}_0\|_2^2 = 1$. Since $\frac{dR_{SK}}{dP}$, the optimal P is P_A . Then, consider the function

$$f(P_A) = \frac{(P_Bx_0 + \sigma^2)(P_Ax_0 + \sigma^2)}{(P_A\sigma^2 + P_B\sigma^2)x_0 + \sigma^4}, \quad (46)$$

where $x_0 = \beta_{ba}\mathbf{w}_0^H\mathbf{J}_A\mathbf{w}_0$.

To analyze the effect of the maximum power budget P_A , the derivation of (46) with respect to P_A yields

$$\begin{aligned} \frac{df(P_A)}{dP_A} &= \frac{x_0(P_Bx_0 + \sigma^2)(P_A\sigma^2x_0 + P_B\sigma^2x_0 + \sigma^4)}{(P_A\sigma^2x_0 + P_B\sigma^2x_0 + \sigma^4)^2} \\ &\quad - \frac{\sigma^2x_0(P_Bx_0 + \sigma^2)(P_Ax_0 + \sigma^2)}{(P_A\sigma^2x_0 + P_B\sigma^2x_0 + \sigma^4)^2} \\ &= \frac{(P_Bx_0 + \sigma^2)P_B\sigma^2x_0^2}{\sigma^4(P_Ax_0 + P_Bx_0 + \sigma^2)^2} \\ &= \frac{(P_Bx_0 + \sigma^2)P_Bx_0^2}{\sigma^2(P_Ax_0 + P_Bx_0 + \sigma^2)^2}. \end{aligned} \quad (47)$$

As P_A gradually increases, the denominator of (47) gradually increases, eventually forcing $\frac{df(P_A)}{dP_A}$ approaches zero such that the value of R_{SK} becomes saturated. This completes the proof.

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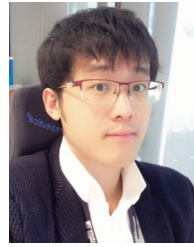
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